

HARCO **INSTALLATION INSTRUCTIONS**

FITTINGS

FITTING to VALVE to PIPE (STYLE A)

2" & 2½" Restraint Assembly

Items to be assembled:

- A - (1) Grip rings w/ clamp bolts & nuts.
- B - (2) I-Bolts or Corten threaded rod & Nuts
- C - (1) Push joint valve.

Step 1: Install valve in system with pipe ends homed to pipe stops.

Step 2: Assemble split ring around socket of bell so it seats up against the backside of valve gasket O.D. The "BOTTOM" ring half is installed on the bottom of the valve, while the other half goes on from the top side.

Step 3: Assemble HARCO grip ring on pipe with the rings flush against valve bells. (as shown in diagram) After being sure of the alignment of the restraint bolt holes, tighten grip ring clamp bolts to 100 ft-lbs. of torque.

Step 4: Slide T-Bolt or Corten threaded rod through ring holes, from grip ring side, thread on Corten nuts tighten evenly to 5 ft-lbs. of torque for a snug fit.

WARNING:

If grip rings are not installed flush against valve bells, the thrust force from valve closure may cause the valve to force it self over end of PVC pipe resulting in failure of gasket seal causing a leak.

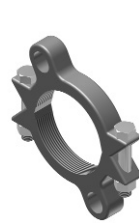


FIGURE 1
(1) GRIP RING w/
BOLTS & NUTS



FIGURE 2
(2) ROD & (4) NUTS

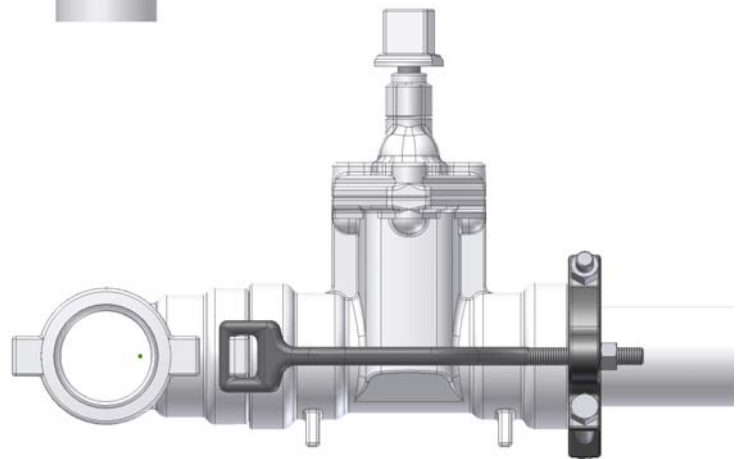
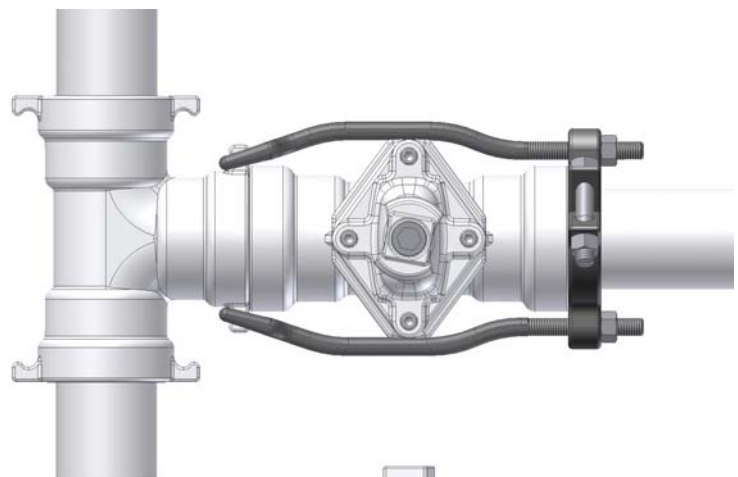
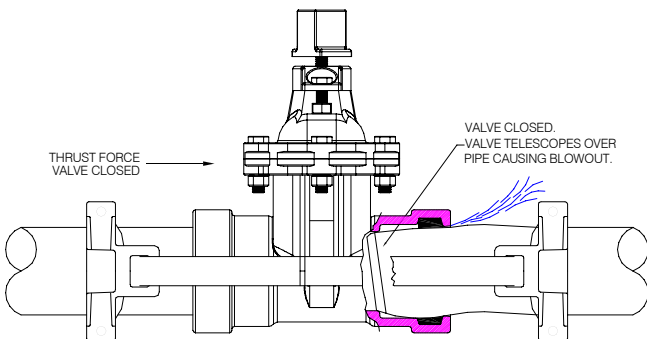


FIGURE 3
ASSEMBLED RESTRAINT



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HARCO **INSTALLATION INSTRUCTIONS**

FITTINGS

FITTING to VALVE to PIPE (STYLE B)

3" - 8" Restraint Assembly

Items to be assembled:

- A - (2) Valve back-up rings consisting of two halves "TOP" (has small notch), "BOTTOM" (has large slot).
- B - (2) HARCO I-Bolts with two low alloy nuts.
- C - (1) Grip ring with clamp bolts and nuts.
- D - (2) T-Bolts and nuts.
- E - (1) Push-on joint valve.
- F - (1) HARCO fitting with lugs.

Notes:

- HARCO recommends all assembly be done prior to placement in the ditch, although assembly in the ditch is possible.
- Not all valve brands and models require all slots provided on split rings.
- I-Bolt is custom length for valve to fitting restraint assembly.

Step 1: Cut IPS-PVC pipe to a length which allows valve and HARCO fitting bells to contact flush against one another (pre-measuring valve and fitting bell depths is required).

Step 2: Insert pipe into valve bell first, then install HARCO fitting onto opposite end, making sure bell ends contact each other as shown.

Step 3: Assemble valve back-up ring around socket of bell so it seats against the valve "gasket race." The "BOTTOM" ring half is installed on the bottom of the valve, while the other half goes on from the top side.

Step 4: Slide HARCO I-Bolt through ring holes on either side, then hook the I-bolts over lugs on the HARCO fitting. Thread Corten nuts on to I-Bolts and tighten until bells are snug.

Step 5: Insert pipe into opposite valve bell.

Step 6: Repeat "Step 3."

Step 7: Clamp grip rings on pipe with the rings flush against valve bells (as shown in diagram). After being sure of the alignment of the restraint bolt holes, tighten grip ring clamp bolts to 100 foot-lbs. of torque.

Step 8: Slide T-Bolt through ring holes, from clamp ring side, thread on Corten nuts tighten evenly to 5 foot-lbs. of torque for a snug fit.

WARNING:

If grip rings are not installed flush against valve bells, the thrust force from valve closure may cause the valve to force it self over end of PVC pipe resulting in failure of gasket seal causing a leak.



FIGURE 4
(2) VALVE (BUR) RING

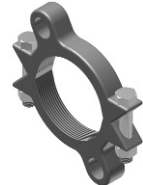


FIGURE 5
(2) GRIP RING w/
BOLTS & NUTS



FIGURE 6
(2) EYEBOLT & NUT



FIGURE 7
(4) T-BOLT & NUT

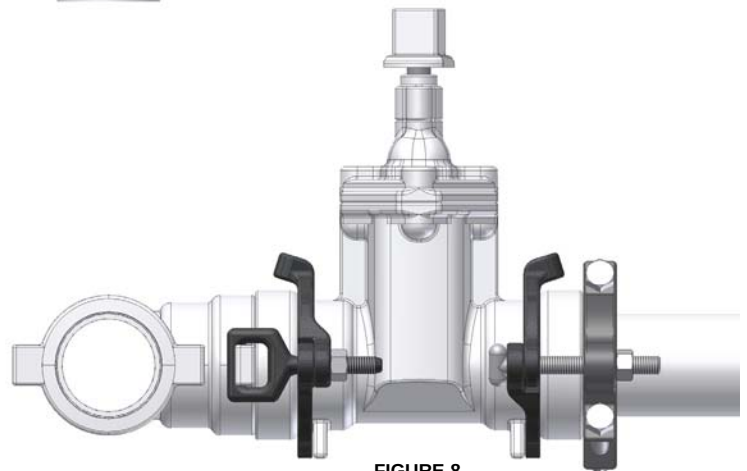
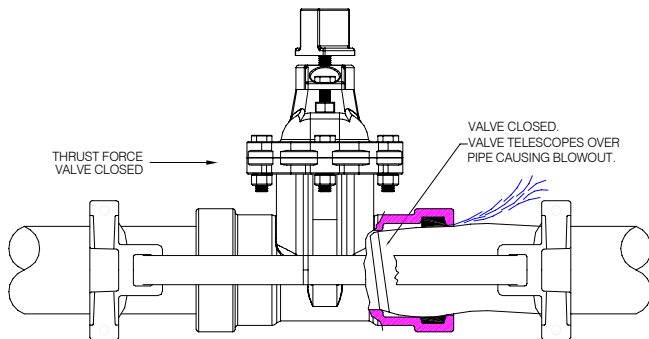
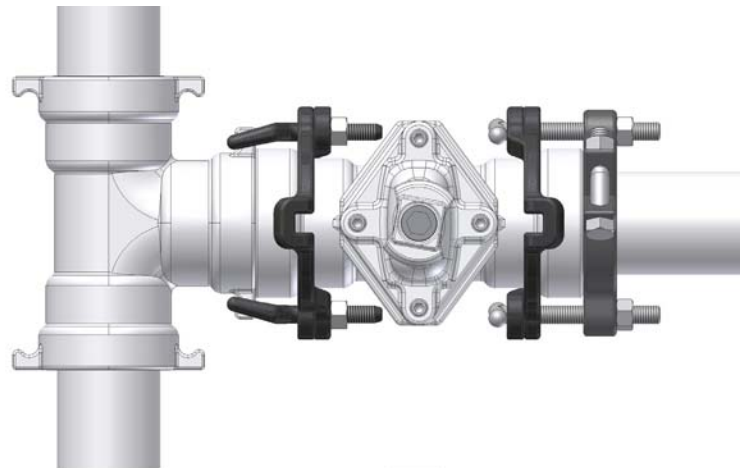


FIGURE 8
ASSEMBLED RESTRAINT



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HARCO **INSTALLATION INSTRUCTIONS**

FITTINGS

FITTING to VALVE to PIPE (STYLE B) **10" & 12" Restraint Assembly**

Items to be assembled:

- A - (1) Grip rings w/ clamp bolts & nuts.
- B - (2) DI restraint rods.
- C - (4) T-bolts & nuts.

Step 1: Install valve in system with short pipe between valve & fitting that seals and allows valve to butt against fitting as shown.

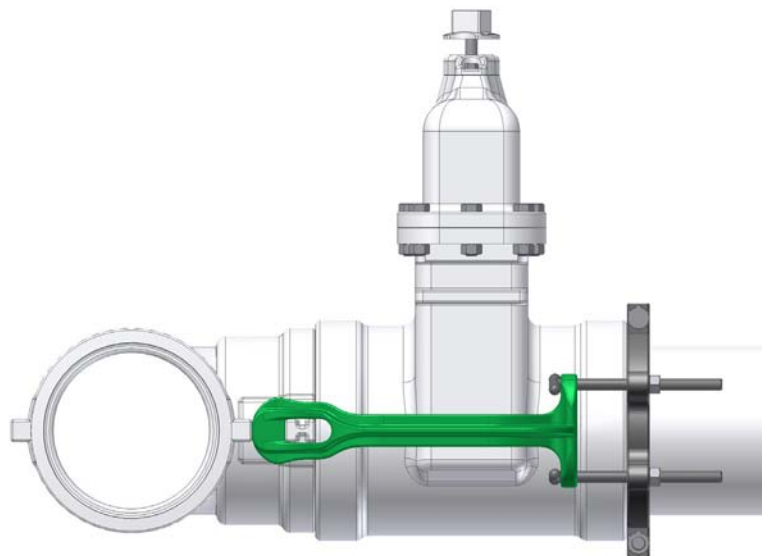
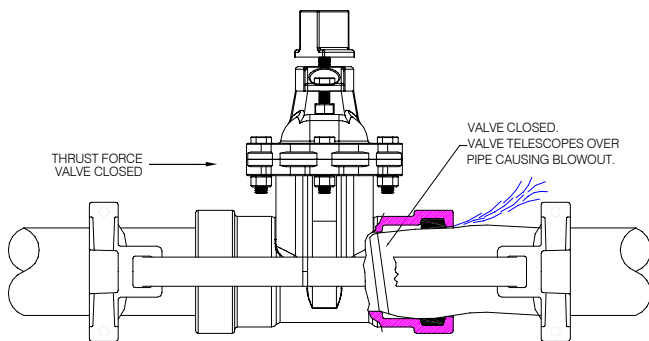
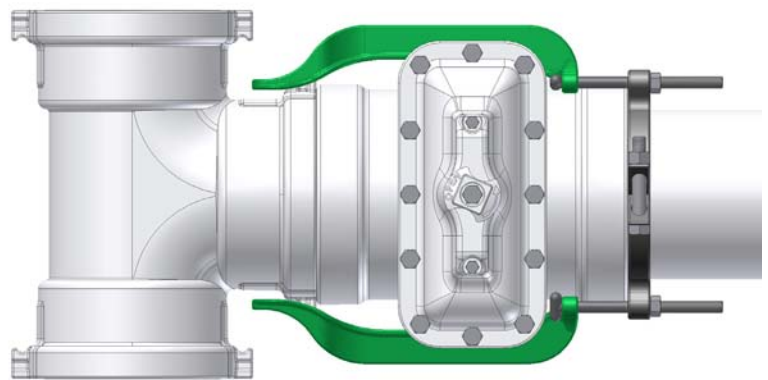
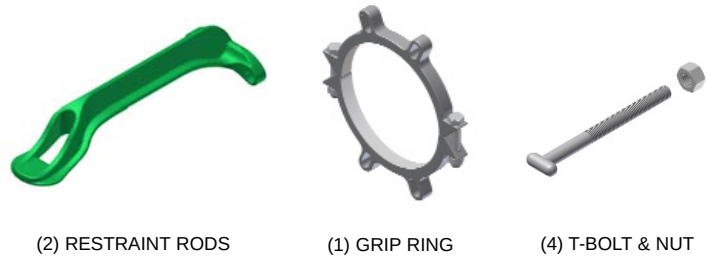
Step 2: Clamp grip rings on pipe with the rings flush against valve bells. (as shown in diagram) Tighten grip ring clamp bolts to 100 ft-lbs. of torque.

Step 3: Start T-Bolts through DI Restraint Rod holes as shown. Then handling this assembly start the T-Bolts through the receiving holes in the grip ring while maneuvering the "eye" at the other end of the DI restraint rod over the lug on the bell of the fitting.

Step 4: Tighten Corten nuts evenly on T-bolts to 5 ft-lbs. of torque for a snug fit.

WARNING:

If grip rings are not installed flush against valve bells, the thrust force from valve closure may cause the valve to force it self over end of PVC pipe resulting in failure of gasket seal causing a leak.



ASSEMBLED RESTRAINT



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